

Lecture 02 — Study Notes

Contradiction and Induction

MIT 6.1200J / 18.062J · Spring 2024

Logical Deduction

This is the **third and most important** component of a proof. It deals with the *structure* of the argument — how each step connects to the next.

Definition — Inference Rule

An **inference rule** is a rule for combining true propositions to form other true propositions.

An inference rule is the machine that drives every step of a proof forward.

Key Inference Rules

Every rule below can be verified with a truth table.

Modus Ponens — “the mode of putting”

If $P \Rightarrow Q$ is true, and P is true, then Q must be true.

$$\frac{P \Rightarrow Q; P}{Q}$$

Written equivalently as: $((P \Rightarrow Q) \text{ and } P) \Rightarrow Q$, or $P \Rightarrow Q, P \vdash Q$.

Modus Tollens — “the mode of taking”

$((P \Rightarrow Q) \text{ and not } Q) \Rightarrow \text{not } P$

If the conclusion is false, the hypothesis must also be false. This is just the contrapositive applied as an inference rule.

Hypothetical Syllogism (chain rule)

$((P \Rightarrow Q) \text{ and } (Q \Rightarrow R)) \Rightarrow (P \Rightarrow R)$

Implications chain: if P leads to Q , and Q leads to R , then P leads to R .

Proof by Contradiction Rule
 $((\text{not } P) \Rightarrow \text{false}) \Rightarrow P$
If assuming NOT P leads to a contradiction, then P must be true. This is the formal basis for the entire contradiction technique.

Proof Standards for This Class

Rule	What it means
Clear steps	Every step must follow logically from the previous
Cite prior work	State which already-proved propositions you are using
No leaps	Do not skip steps or assume the reader “sees it”
No intimidation	“ P is obvious” or “Clearly Q ” is not a proof
No circular	“I already knew P , so it’s an axiom” is invalid when asked to prove P

Note
You do NOT need to cite an inference rule at every step, and you do NOT need to specify precisely which axioms you use — basic math from before this class is fair game. The key is logical clarity, not formalism overload.

Fundamental Proof Techniques

Proving Existence

To prove $\exists x \in S \ P(x)$, find **one concrete example** that works.

Template: Existence Proof
Proof. We’ll show that $x = [\textit{specific value}]$ works. Indeed, for this choice of x , $P(x)$ is true because $[\textit{reasons}]$. $\square$

Example: Prove $\exists n \in \mathbb{N} \ n \geq 10$ and $\text{isPrime}(n)$.
Proof. $n = 17$ works: 17 is prime and $17 \geq 10$. $\square$

Key Takeaway
For existence proofs, one valid example is a **complete** proof. You do not need to find all examples, just one.

Proving Universality

To prove $\forall x \in S P(x)$, you cannot just check examples. Instead: introduce a **generic, arbitrary** element $x \in S$ — one about which you make **no special assumptions** — and prove $P(x)$ holds for it.

Template: Universal Proof

Proof. Assume x is an arbitrary element of S . Then $P(x)$ is true because *[reasons]*. $\square$

Example: Prove $\forall x \in \mathbb{R} x^2 - 6x > -10$.

Proof. Let x be an arbitrary real number. Then:

$$\begin{aligned} x^2 - 6x + 9 &= (x - 3)^2 \geq 0 \\ \therefore x^2 - 6x &= (x - 3)^2 - 9 \geq -9 > -10 \quad \square \end{aligned}$$

Note

The word **arbitrary** is key. It means: “I pick x knowing only that $x \in S$.” Any proof that works for an arbitrary x works for every x .

Proof of an Implication: Direct Method

To prove $P \Rightarrow Q$: **assume P is true**, then **derive Q** .

Template: Direct Proof

Proof. Assume P is true. Then Q is also true because *[reasons]*. $\square$

Example: If n is a multiple of 10, then n is a multiple of 2.

Proof. Assume $n = 10k$ for some integer k . Then $n = 2(5k)$, so n is 2 times the integer $5k$, i.e. a multiple of 2. $\square$

Proof of an Implication: Contrapositive Method

$P \Rightarrow Q$ is **logically equivalent** to $(\text{not } Q) \Rightarrow (\text{not } P)$. When the contrapositive is easier to prove directly, use it.

Template: Contrapositive Proof

Proof. We prove the contrapositive. Assume $\text{not } Q$ is true. Then $\text{not } P$ is true because *[reasons]*. $\square$

Example: For integer n : $(n^2 \text{ is even}) \Rightarrow (n \text{ is even})$.

Proof. Prove contrapositive: $(n \text{ is odd}) \Rightarrow (n^2 \text{ is odd})$.

Assume $n = 2k + 1$. Then:

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$$

which is odd. $\square$

Key Takeaway

When to use contrapositive: when the original “assume P , derive Q ” direction is hard to see, but “assume not Q , derive not P ” is natural. Both prove the same thing.

Proof by Contradiction

Definition — Proof by Contradiction

To prove P , assume not P (the **opposite** of what you want to prove), and derive a **logical impossibility** — a statement that is always false (a **contradiction**). Since the only thing you assumed was not P , the assumption must be wrong, so P must be true.

Formally: prove $(\text{not } P) \Rightarrow \text{false}$. By the inference rule, this gives P .

Also called an **indirect proof**.

The Classic Example: $\sqrt{2}$ is Irrational

Theorem. $\sqrt{2} \notin \mathbb{Q}$ (i.e. $\sqrt{2}$ is irrational).

Proof by Contradiction

Step 1 — Assume the opposite.

Suppose $\sqrt{2} \in \mathbb{Q}$. Then we can write $\sqrt{2} = \frac{a}{b}$ where $a, b \in \mathbb{Z}$ and $\gcd(a, b) = 1$ (fraction in lowest terms).

Step 2 — Derive consequences.

$$\frac{a}{b} = \sqrt{2} \Rightarrow a = b\sqrt{2} \Rightarrow a^2 = 2b^2$$

So a^2 is even. By our lemma (even square $\Rightarrow$ even root), a itself is even. Write $a = 2c$ for some integer c .

Step 3 — Push further.

$$a^2 = 2b^2 \Rightarrow (2c)^2 = 2b^2 \Rightarrow 4c^2 = 2b^2 \Rightarrow 2c^2 = b^2$$

So b^2 is even, meaning b is also even.

Step 4 — Contradiction.

Both a and b are even, meaning they share a factor of 2. But we assumed $\gcd(a, b) = 1$.

Contradiction ($\Rightarrow \leq$ or $\perp$). $\square$

Key Takeaway

The structure is always the same:

1. Assume NOT P
2. Do valid algebra/logic from that assumption
3. Arrive at something that is **provably false**
4. Conclude P must be true

The contradiction does the work — you don't need to “find” truth directly.

Note

A useful lemma used above (and worth remembering): **If n^2 is even, then n is even.**
 Proof via contrapositive: if n is odd, n^2 is odd (shown in §2.4).

Proof Outlining

Before writing a proof, **outline its structure first**. The form of the theorem mechanically tells you what your proof needs to contain — even before you understand the specific mathematical content.

Common Mistake

The most common mistake: diving into **how** to prove something before figuring out **what** you actually need to prove. Outlining prevents this.

Mechanical Decomposition Rules

Theorem form	What your proof must contain
$\forall n \in S \ P(n)$	Introduce arbitrary $n \in S$, then prove $P(n)$
$\exists n \in S \ P(n)$	Find a specific n and verify $P(n)$
$P \Rightarrow Q$	Assume P , derive Q — or prove contrapositive
$P \Leftrightarrow Q$	Two sub-proofs: $(P \Rightarrow Q)$ and $(Q \Rightarrow P)$
not P	Consider contradiction: assume P , derive false

Example Outline (with Nonsense Terms)

Theorem: “For every integer n , n is fooish iff $n + 1$ is barsome.”

Notice: we have $\forall n \in \mathbb{Z}$ and an iff. The outline writes itself:

Outline Step 1 — handle the $\forall$:

Suppose n is any integer; must prove $F(n) \Leftrightarrow B(n+1)$.

Outline Step 2 — handle the iff:

Must show $F(n) \Rightarrow B(n+1)$ and $B(n+1) \Rightarrow F(n)$.

Outline Step 3 — handle each implication:

- To prove $F(n) \Rightarrow B(n+1)$: assume $F(n)$. [TODO: derive $B(n+1)$.]
- To prove $B(n+1) \Rightarrow F(n)$: assume $B(n+1)$. [TODO: derive $F(n)$.]

Key Takeaway

You can build this entire skeleton without knowing what “fooish” or “barsome” mean. The **structure** of a proof is determined by the **logical form** of the statement, not its mathematical content.

Proof by Induction

The most common and most powerful proof technique in computer science. Used whenever you need to prove a statement holds for **all natural numbers** (or all integers $\geq k$, or all elements of a recursively defined set).

The Induction Axiom

Axiom — Mathematical Induction

Let $P(n)$ be a predicate defined for $n \in \mathbb{N}$.

If both of the following hold:

- **Base case:** $P(0)$ is true
- **Inductive step:** $\forall n \in \mathbb{N}, P(n) \Rightarrow P(n+1)$

Then: $\forall n \in \mathbb{N}, P(n)$ is true.

Why Induction Works — Intuition

Think of an infinite chain of dominoes:

You know:

- The first domino falls (base case: $P(0)$)
- Each falling domino knocks over the next (inductive step: $P(n) \Rightarrow P(n+1)$)

Conclusion: **all** dominoes fall ($\forall n, P(n)$).

Formally, from $P(0)$ and $P(0) \Rightarrow P(1)$, we get $P(1)$. From $P(1)$ and $P(1) \Rightarrow P(2)$, we get $P(2)$. And so on forever. The induction axiom packages this infinite chain into one finite statement.

Standard Template

Template: Proof by Induction

Proof. Let $P(n)$ be [state the predicate]. We prove $\forall n \in \mathbb{N}, P(n)$ by induction on n .

Base case ($n = 0$): [verify $P(0)$ directly]. $\square$

Inductive step: Let $n \in \mathbb{N}$ and assume $P(n)$ is true (this assumption is the **induction hypothesis**). We must show $P(n + 1)$.

[algebra/logic using the induction hypothesis]

Therefore $P(n + 1)$ holds. By induction, $P(n)$ is true for all $n \in \mathbb{N}$. $\square$

Worked Example: Gauss Sum Formula

Theorem. $\forall n \in \mathbb{N}, 1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$

Full Induction Proof

Let $P(n)$ be: $1 + 2 + \dots + n = \frac{n(n+1)}{2}$.

Base case ($n = 0$): LHS = 0, RHS = $(0)\frac{1}{2} = 0$. Equal. $\checkmark$

Inductive step: Assume $P(n)$: $1 + 2 + \dots + n = \frac{n(n+1)}{2}$. Show $P(n + 1)$: $1 + 2 + \dots + n + (n + 1) = (n + 1)\frac{n+2}{2}$.

Starting from the LHS of $P(n + 1)$:

$$(1 + 2 + \dots + n) + (n + 1) = \frac{n(n+1)}{2} + (n + 1)$$

(used IH on the parenthesised sum)

$$= (n + 1) \cdot \left(\frac{n}{2} + 1 \right) = (n + 1) \cdot \frac{n + 2}{2}$$

This is exactly $(n + 1)\frac{n+2}{2}$, which is the RHS of $P(n + 1)$. $\checkmark$

By induction, $P(n)$ holds for all $n \in \mathbb{N}$. $\square$

Strengthening the Induction Hypothesis

Sometimes the predicate you set out to prove is **too weak** to induct on. The inductive step can't go through because the hypothesis doesn't give you enough to work with.

Counterintuitive fix: prove a **stronger** statement. A stronger $P(n)$ is harder to establish in the base case, but gives you **more to assume** in the inductive step — which often more than compensates.

The L-Tromino Example

Setting: A $2^n \times 2^n$ grid. Can you tile it with L-shaped trominoes, leaving exactly one center-most square uncovered?

Weak predicate $P(n)$:

“Can tile $2^n \times 2^n$ grid leaving one **center** square uncovered.”

Problem: when doing the inductive step, knowing you can tile a 2^n grid with the center missing doesn't directly help you tile the 2^{n+1} grid. The four sub-grids' “missing squares” are not the center of the big grid.

Stronger predicate $Q(n)$:

“Can tile $2^n \times 2^n$ grid leaving **any one** square uncovered.”

$Q(n)$ is strictly stronger: it covers all possible missing positions. But in the inductive step, you can now place one tromino at the center of the 2^{n+1} grid and apply $Q(n)$ to each quadrant. The proof works!

Inductive Step Sketch for $Q(n)$

Assume $Q(n)$. Given a $2^{n+1} \times 2^{n+1}$ grid, suppose square (i, j) must be left uncovered. WLOG (i, j) is in the top-left quadrant.

- Place one L-tromino at the center of the big grid, covering one square from each of the three quadrants **not** containing (i, j) .
- Apply $Q(n)$ to the top-left quadrant (leaving (i, j) uncovered).
- Apply $Q(n)$ to each of the other three quadrants (leaving the tromino-covered square uncovered in each). ✓

Since $Q(n) \Rightarrow P(n)$ trivially, $\forall n, P(n)$ follows. $\square$

Key Takeaway

Takeaway: When induction fails, try strengthening the hypothesis.

More to assume in the inductive step can be worth more than less to prove in the base case. Finding the right IH is part skill, part intuition — it gets easier with practice.

A bonus: stronger induction hypotheses often yield **constructive** proofs (they produce an algorithm, not just an existence result).

Quick Reference — All Four Techniques

Technique	Prove	Strategy	Signal
-----------	-------	----------	--------

Direct	$P \Rightarrow Q$	Assume P , derive Q	Clear forward path
Contrapositive	$P \Rightarrow Q$	Assume not Q , derive not P	Forward is murky
Contradiction	any P	Assume not P , derive false	Negation has useful structure
Induction	$\forall n \in \mathbb{N}, P(n)$	Base case + $P(n) \Rightarrow P(n+1)$	Statement involves all of $\mathbb{N}$

Common Mistakes

- **Proof by example for universals** — one or a million examples never proves $\forall n, P(n)$. Only induction or a generic-element argument works.
- **Confusing contrapositive with converse** — $(P \Rightarrow Q)$ is equivalent to $(\text{not } Q \Rightarrow \text{not } P)$, NOT to $(Q \Rightarrow P)$. Using the converse proves nothing.
- **Circular contradiction** — reaching “not P ” from “assume not P ” is not a contradiction; you just restated your assumption.
- **Forgetting the base case in induction** — without it, the chain never starts. The inductive step alone proves nothing.
- **Using $P(n)$ in the inductive step before establishing it** — the induction hypothesis gives you $P(n)$; you must show $P(n+1)$, not just wave your hands at “ $P(n+1)$ follows similarly”.
- **Weak induction hypothesis** — if the step gets stuck, the IH may be too weak. Strengthen it (prove more) rather than looking for clever tricks.